

Simulation study design for investigating Random Forests for Variable Importance

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1 Purpose of simulation study

The purpose of the present simulation study is to investigate the performance of various variable importance measures for random forests with respect to identifying effects in a sparse setting.

2 Simulation setting

We will simulate from a logistic regression model with a single binary outcome, Y and 200 covariates, $(X_1, \dots, X_{200} = X)$.

$$EY = \text{logis}(\theta_0 + \theta X) \tag{1}$$

The vector of effects θ will be sparse: only 10 entries will be non-zero.

3 Parameters of simulation

We will investigate a range of sample sizes to assess performance in different application settings. The grid of sample sizes will be $10^3, 10^4, 10^5, 10^6$.

Each method under investigation will have its hyperparameters tuned. The hyperparameter set with least out-of-bag/cross-validation error will be used in the final comparison between methods.

4 Variable importance measures

For random forests we will consider permutation-based variable importance as well as minimal depth as variable importance measure. For a comparison outside the random forest framework, we will use L^1 penalized regression / Dantzig selector.

5 Notions of scoring

Exactly how performance of variable importance measures is to be scored is unclear to me at this time. A non-parametric method of scoring would be preferable, as different notions of importance might not be directly comparable. If each variable importance measure admits a ranking (preferably allowing for ties), the accuracy of importance rankings may be evaluated against a ground truth of importance rank. The task of selecting a subset of important variables from a large set of mostly unimportant variables may be different than ranking the importance of a relatively small set of variables that is known to have some importance. These two tasks may be in tension for some methods. Perhaps it should be further clarified what kind of task we wish to score with respect to.